DSAA3073: Theories in Data Science

Tutorial 6A: Derive AdaBoost One Dependency at a Time

150-minute core tutorial - 15-minute break - 60-minute protected buffer

Wei WANG · HKUST(GZ) · 2026-08-28

Explorative projection

LESSON BACKBONE

Realizable PAC learning asks for arbitrarily small error, but an efficient learner may offer only a fixed advantage over guessing. Which exact objective and update can turn that limited advantage into an ensemble?

We begin by separating two statements that are easy to blur: a population weak-learning guarantee and one observed weighted error on a fixed training sample. Only after that distinction is secure will we assign mass to training examples, choose a loss, and derive the coefficient that changes the mass.

The order is deliberate. The current example distribution must exist before its error can be measured; the error must exist before a coefficient can be chosen; the coefficient must be derived before the update is legitimate. By the end, every line of AdaBoost will have a mathematical reason.

The three core blocks take **45, 45, and 60 minutes**. Take the scheduled 15-minute break after the first block. The final 60 minutes are protected for questions, reconstruction, and recovery; they introduce no required knowledge.

Clock	Mathematical work	Protected time
0-45	Separate weak and strong learning; identify the quantifiers hidden by a one-sample accuracy claim	45 min core
45-60	Scheduled break	15 min
60-105	Construct a normalized training-example distribution and a pointwise exponential upper bound	45 min core
105-165	Derive the log-odds coefficient, normalize one update, and assemble AdaBoost	60 min core
165-225	Questions, recovery, alternative calculations	60 min buffer

A fixed advantage is not yet arbitrary accuracy

Chapter 5 left exactly three earlier ideas available here, in this order. They are enough to pose the boosting problem; no information criterion or model-selection shortcut is needed.

No.	Available idea	Use here
1	realizable PAC learning	State the population accuracy and confidence target when a zero-risk hypothesis is represented.
2	convex-function background	Recognize why a one-dimensional convex objective can be solved from its derivative.
3	empirical risk minimization	Choose a hypothesis that minimizes a declared empirical loss over its class.

IN-CLASS CHECK

Prompt: Reconstruct the three items in order. Which one describes a population guarantee, and which one describes an optimization rule on observed data?

Hint: Do not turn an empirical minimizer into a high-probability guarantee without a theorem.

Answer:

> The order is realizable PAC learning, convex-function background, and ERM. Realizable PAC learning supplies the population guarantee. ERM is the rule that minimizes the chosen empirical loss. Convexity describes objective geometry; neither ERM nor convexity alone supplies the PAC conclusion.

Recall the established notation $\mathcal{H}$ for a hypothesis class, $S = ((x_i, y_i))_{i=1}^m$ for an i.i.d. labeled sample, and (ε, δ) for PAC accuracy and failure probability. In the realizable binary setting, a strong target has the form

$$\Pr_S[R(h_S) \le \varepsilon] \ge 1 - \delta, \tag{1}$$

for every requested $(\varepsilon, \delta) \in (0,1)^2$, with the declared sample and computational resources polynomial in the relevant size parameters. The probability is over the random sample; R is population risk.

Before naming a weaker contract, complete the two cards below. Put **arbitrary $\boldsymbol{\varepsilon}$** on the card that must meet every requested accuracy; put **fixed positive advantage**, **every population distribution in scope**, **confidence $\mathbf{1-\boldsymbol{\delta}}$**, and **polynomial resources** on the other. Then mark 0.49 on the ruler and decide whether one observed 51-percent accuracy alone fills every slot on either card.

Guarantee card A	Guarantee card B
accuracy target:	accuracy target:
distribution quantifier:	distribution quantifier:
confidence:	confidence:
resources:	resources:

error 0		error $\frac{1}{2}$
	mark 0.49, 0.40, and 0.25; write each distance from $\frac{1}{2}$	

Table 1: Two guarantee cards and an error ruler. The labels are withheld until the quantifiers have been placed.

IN-CLASS CHECK

Prompt: Why does 51 percent accuracy on one observed sample fail to establish the weaker card? Name at least three missing quantifiers or controls.

Hint: Ask what varies, what probability is promised, and how resources scale.

Answer:

It supplies only one empirical observation. It does not quantify over every population distribution in the declared realizable scope, does not provide a confidence statement over sample randomness, does not establish a fixed positive advantage uniformly in that scope, and does not state polynomial sample or computational resources.

Definition: Weak learner and weak-learning edge

In a declared realizable binary-classification scope, a **weak PAC learner** has some fixed $\gamma > 0$ and, for every allowed population distribution and confidence request δ, returns with probability at least $1 - \delta$ a hypothesis with population error at most

$$\frac{1}{2} - \gamma, \tag{2}$$

using the declared polynomial resources. The positive number γ is the **edge**: the guaranteed advantage over random guessing.

Inside the AdaBoost training loop we will need a related empirical condition: when called on the current normalized weights of a fixed training sample, the base learner returns a binary hypothesis with weighted error below $\frac{1}{2}$. That roundwise observation is not, by itself, the population weak-PAC guarantee above.

If a current error is $e < \frac{1}{2}$, its observed advantage is $\frac{1}{2} - e$. Thus errors 0.49, 0.40, and 0.25 correspond to observed advantages 0.01, 0.10, and 0.25. A later uniform training-rate statement would need a positive lower bound that holds at every relevant round, not just the largest number in this list.

IN-CLASS CHECK

Prompt: A base classifier has weighted training error 0.40 in one round. Compute its observed advantage. Which additional statement would be needed to use $\gamma = 0.10$ uniformly over all rounds, and what population claim still does not follow?

Hint: Separate current-round arithmetic, an all-round training condition, and a population guarantee.

Answer:

The observed advantage is $\frac{1}{2} - 0.40 = 0.10$. A uniform round condition must state that every returned base hypothesis has weighted error at most 0.40 under every example distribution produced in the run. Even then, this empirical condition alone does not prove a weak-PAC guarantee over every population distribution or a test-error guarantee.

Definition: Strong learner

A **strong PAC learner** meets every requested accuracy $\varepsilon > 0$ and confidence level $1 - \delta$ in the declared PAC setting, with the relevant polynomial resource dependence. A fixed error target such as 0.10 is not strong learnability for arbitrary ε.

The two cards can now be labeled. The weak card fixes γ and promises error at most $\frac{1}{2} - \gamma$; the strong card accepts an arbitrary ε. A booster must do more than repeat the same weak classifier: it must make the base learner confront different mistakes while keeping the training sample fixed. That creates the need for a distribution over training indices.

Put objective mass where the current score struggles

The four rows below come from one fixed training sample. Normalize the raw masses first. Then compute the signed score $y_i g(x_i)$, the classification error indicator, and $\exp(-y_i g(x_i))$. Sort the rows by the last column. Before reading a reweighting rule, predict which row a smooth objective will emphasize most.

i	y_i	$g(x_i)$	raw mass	normalized	error indicator	$\exp(-y_i g(x_i))$
1	$+1$	1.0	4			
2	$+1$	-0.5	2			
3	-1	-0.2	1			
4	-1	0.4	3			

IN-CLASS CHECK

Prompt: Complete the normalized-mass column. Which row has the largest smooth-loss value, and why is that answer not simply "the row with largest mass"?

Hint: The raw masses sum to 10; compare signed scores before exponentiating.

Answer:

> The normalized masses are 0.4, 0.2, 0.1, and 0.3. The signed scores are 1.0, -0.5, 0.2, and -0.4, so the smooth-loss values are approximately 0.368, 1.649, 0.819, and 1.492. Row 2 has the largest loss because its signed score is most negative; objective influence depends on both its current mass and its loss, not on mass alone.

Definition: AdaBoost example distribution

At round t, the registered notation $D_{t(i)}$ is the positive normalized weight of training index i:

$$D_{t(i)} > 0, \quad \sum_{i=1}^{m} D_{t(i)} = 1. \tag{3}$$

The base learner is scored on the fixed sample under these weights. D_t is a distribution on the finite index set $\{1, \ldots, m\}$; it is not the unknown population distribution quantified in a PAC guarantee.

IN-CLASS CHECK

Prompt: A learner is perfect on indices carrying total D_t-mass 0.7 and wrong on the rest. What is known, and what is not known?

Hint: Name the sample, the round, and the population separately.

Answer:

> Its current weighted empirical error on this fixed sample is 0.3. We do not yet know its error under another round's weights, its population risk, or a confidence statement over new samples. Those require separate conditions or theorems.

The weights tell us **where** to look, but not how severely to charge a real score. For a binary label $y \in \{-1, +1\}$ and any real score $F(x)$, correctness depends on the sign of $yF(x)$. A smooth objective should remain large when that signed score is negative and decrease as it becomes positive.

Definition: Exponential loss

The registered exponential surrogate loss is

$$\ell_{\text{exp}}(y, F(x)) = e^{-yF(x)}. \tag{4}$$

It is smooth and convex as a function of the signed score. Pointwise,

$$1_{[yF(x)\leq 0]} \leq e^{-yF(x)}. \tag{5}$$

The left side records a classification mistake; the right side is a smooth upper bound, not an observed error rate.

The upper bound needs only two sign cases. If $yF(x) \leq 0$, the indicator equals 1 and $e^{-yF(x)} \geq 1$. If $yF(x) > 0$, the indicator equals 0 and the exponential is positive. No approximation is involved.

IN-CLASS CHECK

Prompt: Verify the pointwise bound at signed scores -0.4, 0, and 1.2. At which value is the indicator discontinuous while the surrogate remains smooth?

Hint: Compute the indicator and exponential separately.

Answer:

> At -0.4, the values are 1 and $e^{0.4} \approx 1.492$; at 0, they are 1 and 1; at 1.2, they are 0 and $e^{-1.2} \approx 0.301$. The indicator jumps as the signed score crosses zero, while the exponential loss is smooth everywhere.

Fix a binary candidate hypothesis h for the current round and consider adding it to the current score with a scalar coefficient a. Once the current state has been absorbed into D_t, the new one-dimensional objective is proportional to

$$\Phi_{t(a)} = \sum_{i=1}^{m} D_{t(i)} e^{-a y_i h(x_i)}. \tag{6}$$

This function is convex in a. That fact allows exact line minimization; it does not by itself prove a training rate or good population performance.

IN-CLASS CHECK

Prompt: Classify each statement as supported now or not yet supported: (a) Φ_t is convex; (b) minimizing Φ_t guarantees small test error; (c) its minimizer can be found from a derivative under the stated binary setup; (d) convexity alone proves an exponential training-error rate.

Hint: Separate objective geometry from statistical and convergence conclusions.

Answer:

> Statements (a) and (c) are supported. Statements (b) and (d) are not: population performance needs a generalization argument, and the training rate needs the exact normalization-factor identity plus an all-round edge condition. Those results are not available yet.

The objective has now exposed the missing quantity: its minimizer. Only now is it legitimate to name the weighted error that splits correct from incorrect mass and derive the coefficient.

Let correct and incorrect mass balance the coefficient

Before naming the coefficient, take a generic scalar a. Put the current correctly classified mass on the left pan and the error mass on the right. Write the factor multiplying each pan after adding $ah(x)$. Then sketch the two-term curve and mark where its derivative should vanish. In the four-row wheel, leave the final normalized column blank.

correct mass	balance	error mass
$1-e$ times factor	$\Phi_{t(a)}$	e times factor
decreases as a grows	$\Phi'_{t(a)}=0$ at	increases as a grows

i	before	correct?	factor	normalized after
1	0.10	yes		
2	0.25	no		
3	0.25	yes		
4	0.40	yes		

Table 2: Correct-versus-error balance and one unfinished update wheel. The mathematical names arrive only after the split has been derived.

Definition: Weighted error and AdaBoost coefficient

For the round-t binary hypothesis h_t, its registered weighted error is

$$\varepsilon_t = \Pr_{i\sim D_t}\left[h_{t(x_i)} \neq y_i\right] = \sum_{i:h_{t(x_i)}\neq y_i} D_{t(i)}. \tag{7}$$

We assume $0<\varepsilon_t<\frac{1}{2}$. The **AdaBoost weak-hypothesis coefficient** is the exact minimizer of the roundwise exponential objective:

$$\alpha_t = \frac{1}{2}\log\frac{1-\varepsilon_t}{\varepsilon_t}. \tag{8}$$

To derive the formula, split the normalized mass by the sign $y_i h_{t(x_i)} \in \{-1,+1\}$. Correct rows contribute total mass $1-\varepsilon_t$ and factor e^{-a}; incorrect rows contribute ε_t and factor e^a. Therefore

$$\Phi_{t(a)} = (1-\varepsilon_t)e^{-a} + \varepsilon_t e^a. \tag{9}$$

Differentiate:

$$\Phi'_{t(a)} = -(1-\varepsilon_t)e^{-a} + \varepsilon_t e^a. \tag{10}$$

The stationary equation is

$$(1-\varepsilon_t)e^{-a} = \varepsilon_t e^a \quad \leftrightarrow \quad e^{2a} = \frac{1-\varepsilon_t}{\varepsilon_t}. \tag{11}$$

Thus $a=\alpha_t$. Moreover,

$$\Phi''_{t(a)} = (1 - \varepsilon_t)e^{-a} + \varepsilon_t e^{a} > 0, \tag{12}$$

so this is the unique finite minimizer under $0 < \varepsilon_t < \frac{1}{2}$. Because the error is below $\frac{1}{2}$, $\alpha_t > 0$. This exact two-term calculation, not an invented performance trace, determines the coefficient.

IN-CLASS CHECK

Prompt: Derive α_t when $\varepsilon_t = \frac{1}{4}$. What happens to the coefficient as the error approaches $\frac{1}{2}$ from below?

Hint: Use the correct-to-error ratio, then take the limit of its logarithm.

Answer:

> $\alpha_t = \frac{1}{2}\log(3) \approx 0.5493$. As $\varepsilon_t \uparrow \frac{1}{2}$, the ratio $\frac{1-\varepsilon_t}{\varepsilon_t}$ approaches 1, so α_t approaches 0. A near-random hypothesis receives almost no ensemble weight.

Normalize before calling the next round

The same signed factor that defined the objective updates the example distribution. First form unnormalized weights

$$\tilde{D}_{t+1}(i) = D_{t(i)} e^{-\alpha_t y_i h_{t(x_i)}}. \tag{13}$$

Their total is the positive normalization factor

$$Z_t = \sum_{j=1}^{m} D_{t(j)} e^{-\alpha_t y_j h_{t(x_j)}}. \tag{14}$$

Only then define

$$D_{t+1}(i) = \frac{D_{t(i)} e^{-\alpha_t y_i h_{t(x_i)}}}{Z_t}. \tag{15}$$

Summing the last display over i gives 1. A correct row has factor $e^{-\alpha_t} < 1$; an incorrect row has factor $e^{\alpha_t} > 1$. The comparison is relative and becomes a probability distribution only after division by Z_t.

Worked example: One exact four-row update

Use the unfinished wheel: $D_t = (0.10, 0.25, 0.25, 0.40)$ and suppose only row 2 is misclassified. Then $\varepsilon_t = 0.25$ and $\alpha_t = \frac{1}{2}\log 3$. Correct rows receive factor $e^{-\alpha_t} = \frac{1}{\sqrt{3}}$, while row 2 receives $e^{\alpha_t} = \sqrt{3}$.

The unnormalized weights are

$$\left(\frac{0.10}{\sqrt{3}}, 0.25\sqrt{3}, \frac{0.25}{\sqrt{3}}, \frac{0.40}{\sqrt{3}}\right). \tag{16}$$

Their sum is $Z_t = \frac{\sqrt{3}}{2} \approx 0.8660$. Dividing gives

$$D_{t+1} = \left(\frac{1}{15}, \frac{1}{2}, \frac{1}{6}, \frac{4}{15}\right). \tag{17}$$

These values sum to one. The previously incorrect row now carries half the mass; the three correct rows together carry the other half.

IN-CLASS CHECK

Prompt: Recompute the four normalized weights without decimals. Why do total correct and incorrect mass both become $\frac{1}{2}$ in this binary exact-minimization case?

Hint: Use the stationary equality from the derivative calculation.

Answer:

The weights are $\frac{1}{15}, \frac{1}{2}, \frac{1}{6}, \frac{4}{15}$ and sum to 1. At the optimum, $(1-\varepsilon_t)e^{-\alpha_t} = \varepsilon_t e^{\alpha_t}$. These are precisely the two unnormalized group masses, so normalization assigns each group half of the next distribution's total mass. Individual rows within a group retain their relative previous weights.

The loop now has no unexplained line

Definition: AdaBoost

For binary labels and binary base hypotheses, AdaBoost starts from the uniform distribution $D_1(i) = \frac{1}{m}$. At each round $t = 1, ..., T$ it:

1. obtains a base hypothesis h_t using the current D_t;
2. measures $\varepsilon_t = \Pr_{i \sim D_t}\left[h_{t(x_i)} \neq y_i\right]$ and requires $0 < \varepsilon_t < \frac{1}{2}$ in this derivation;
3. sets $\alpha_t = \frac{1}{2} \log\left(\frac{1-\varepsilon_t}{\varepsilon_t}\right)$;
4. computes Z_t and the normalized update $D_{t+1}(i) = D_{t(i)} \frac{e^{-\alpha_t y_i h_{t(x_i)}}}{Z_t}$.

Its registered real-valued ensemble score is

$$F_{T(x)} = \sum_{t=1}^{T} \alpha_t h_{t(x)}, \tag{18}$$

and its binary prediction is $\operatorname{sign}\left(F_{T(x)}\right)$. The score and its sign are different mathematical objects.

The score accumulates evidence; the sign converts that real number into a label. No theorem in this Tutorial says that an arbitrary sequence of such updates has small test error. Tutorial 6B will instead begin with a narrower question: how do the normalization factors control empirical classification error? A formal confidence normalization, its generalization theorem, and a coordinate-optimization interpretation are deferred to Chapter 7.

IN-CLASS CHECK

Prompt: Audit this proposed loop: initialize D_1 uniformly; choose h_t; set $\alpha_t = 1 - \varepsilon_t$; update without Z_t; return F_T as the predicted label. Identify every error and repair it in temporal order.

Hint: Check the error definition, coefficient, normalization, and output type.

Answer:

> First measure ε_t under the already normalized D_t. Replace the linear weight by $\alpha_t = \frac{1}{2}\log\left(\frac{1-\varepsilon_t}{\varepsilon_t}\right)$. Form Z_t from the unnormalized signed exponential factors and divide by it to obtain D_{t+1}. Finally return F_T as a real score and $\operatorname{sign}(F_T)$ as the predicted label. The next round may use only the normalized D_{t+1}.

Standard language to carry forward

Term or notation	Meaning here	Recall cue
weak learner	A learner with a fixed positive edge in its declared guarantee; in the training loop, a base learner must beat one half under current weights.	$\gamma > 0$; keep population and roundwise statements distinct
strong learner	A PAC learner that meets arbitrary requested accuracy and confidence levels with declared polynomial resources.	arbitrary (ε, δ)
example distribution	A normalized positive distribution on fixed training indices at round t.	$D_{t(i)} > 0$, $\sum_i D_{t(i)} = 1$
exponential loss	A smooth convex upper bound on binary classification error as a function of signed score.	$\ell_{\text{exp}}(y, F(x)) = e^{-yF(x)}$
weighted error	The current base hypothesis's error measured under D_t.	$\varepsilon_t = \Pr_{i\sim D_t}\left[h_{t(x_i)} \neq y_i\right]$
AdaBoost coefficient	The exact minimizer of the binary roundwise two-term exponential objective.	$\alpha_t = \frac{1}{2}\log\left(\frac{1-\varepsilon_t}{\varepsilon_t}\right)$
ensemble score	The accumulated real score before a sign is taken.	$F_{T(x)} = \sum_t \alpha_t h_{t(x)}$

Reconstruct the exact Tutorial 6B entry

Tutorial 6B may use exactly the following fifteen items, in order. The first three pass through from earlier chapters; the remaining twelve have now been defined, derived, or decoded here.

No.	Available item	What 6B may ask you to do
1	realizable PAC learning	Distinguish a population guarantee from an empirical bound.
2	convex-function background	Recognize the objective geometry already used in the line minimization.
3	empirical risk minimization	Interpret a weighted base-hypothesis choice as an empirical optimization step.
4	weak learner	State its quantified role and the separate roundwise condition.
5	edge	Express advantage over one half and audit whether it holds round by round.
6	strong learner	Identify the arbitrary-accuracy target of boosting.
7	example distribution	Compute and interpret normalized mass on training indices.
8	exponential loss	Use the pointwise upper bound on classification error.
9	weak-hypothesis coefficient	Derive the two-term minimum from the current weighted error.
10	AdaBoost	Execute the normalized round and distinguish score from prediction.
11	$D_{t(i)}$	Read the current training-index weight.
12	$\varepsilon_t = \Pr_{i\sim D_t}\left[h_{t(x_i)} \neq y_i\right]$	Measure the round's weighted error only after D_t is known.
13	$\alpha_t = \frac{1}{2}\log\left(\frac{1-\varepsilon_t}{\varepsilon_t}\right)$	Assign the exact log-odds coefficient.

No.	Available item	What 6B may ask you to do
14	$F_{T(x)} = \sum_{t=1}^{T} \alpha_t h_{t(x)}$	Use the real ensemble score before taking its sign.
15	$\ell_{\exp}(y, F(x)) = e^{-yF(x)}$	Relate signed score to the smooth pointwise upper bound.

IN-CLASS CHECK

Prompt: Without looking back, reconstruct all fifteen items in order. Then write one sentence explaining why a formal empirical training-error theorem, a normalized confidence definition, and coordinate descent are not on the list.

Hint: Group the list as three pass-through ideas, seven concepts, and five registered notation items.

Answer:

The order is: realizable PAC learning; convex-function background; ERM; weak learner; edge; strong learner; example distribution; exponential loss; weak-hypothesis coefficient; AdaBoost; $D^{(t)}$; ε_t; α_t; $F_{T(x)}$; and exponential-loss notation. The empirical training-error theorem is the next Tutorial's result, while the formal normalized-confidence and coordinate-descent objects are deferred to Chapter 7; none may be used as an input here.

Source note

The historical Week 6 Learning Sheet fixes the official Tutorial 6A scope on printed/PDF pp. 1-20. Its one-dataset performance stories, heuristic reweighting numbers, and any move from one observed 51-percent accuracy to a distribution-uniform guarantee are not controlling evidence and are not reused. No invented test-error curve or performance equivalence appears here.

The population weak-learning definition, weak-to-strong objective, AdaBoost pseudocode, normalized example distribution, signed update, coefficient, and score follow Mohri, Rostamizadeh, and Talwalkar, Sections 7.1-7.2.2, printed pp. 145-154 / PDF pp. 162-171. The strong-versus-weak accuracy and computational distinction, weighted empirical base-hypothesis call, and training-score construction are checked against Shalev-Shwartz and Ben-David, Sections 10.1-10.2, printed/PDF pp. 131-137.

The exponential surrogate and its limitation follow Schapire, **Explaining AdaBoost**, Section 4, printed/PDF pp. 7-9. Smooth convex loss minimization alone does not establish AdaBoost's training rate or good population performance. The product training bound is reserved for Tutorial 6B; the formal confidence normalization, its generalization theorem, and the coordinate-descent interpretation are deferred to Chapter 7.

DSAA3073: Theories in Data Science

Tutorial 6B: AdaBoost Guarantees and Their Limits

150-minute core tutorial - 15-minute break - 60-minute protected buffer

Wei WANG · HKUST(GZ) · 2026-08-28

Explorative projection

LESSON BACKBONE

AdaBoost's update is available. Which exact identity turns its round-by-round normalization into a training-error guarantee, and what remains unproved after that guarantee becomes small?

Tutorial 6A explained every line of the algorithm. Here we make those lines pay for a theorem. We will first follow one training weight through all rounds and show why the normalization factors multiply. Only then will we replace each factor by an edge-dependent upper bound.

A small training-error certificate is useful, but it does not freeze the real ensemble scores and it does not certify test behavior. The second half of the lesson therefore asks what can still change after every training sign is correct, then separates three ways of reading the same AdaBoost state without pretending that they prove the same conclusion.

The three core blocks take **60, 45, and 45 minutes**. Take the scheduled 15-minute break after the first block. The final 60 minutes are protected for questions, proof reconstruction, and recovery; they introduce no required knowledge.

Clock	Mathematical work	Protected time
0-60	Derive the exact product identity, the varying-edge bound, and the fixed-edge corollary	60 min core
60-75	Scheduled break	15 min
75-120	Diagnose what zero empirical error discards and qualify possible later test improvement	45 min core
120-165	Match algorithmic, optimization, and signed-score views to the claims they support	45 min core
165-225	Questions, recovery, alternative proof routes	60 min buffer

Make every normalization factor pay for the final error

Tutorial 6A made exactly fifteen items available, in this order. The first three pass through from earlier chapters; the next seven are established concepts; the last five are the registered forms used in the calculation below.

No.	Available item	Use in this Tutorial
1	realizable PAC learning	Keep a population claim distinct from the empirical theorem proved here.
2	convex-function background	Recognize the roundwise objective geometry without calling it the rate proof.
3	empirical risk minimization	Interpret the weighted base-hypothesis choice on observed data.
4	weak learner	Supply the declared advantage condition.
5	edge	Write $\gamma_t = \frac{1}{2} - \varepsilon_t$ round by round.
6	strong learner	Recall the arbitrary-accuracy goal that motivates boosting.
7	example distribution	Use normalized training-index weights.
8	exponential loss	Upper-bound the zero-one indicator pointwise.
9	weak-hypothesis coefficient	Use the exact log-odds minimizer.
10	AdaBoost	Unroll the normalized update.
11	$D_{t(i)}$	Track one training index through the rounds.
12	$\varepsilon_t = \Pr_{i\sim D_t}\left[h_{t(x_i)} \neq y_i\right]$	Split correct and incorrect mass.
13	$\alpha_t = \frac{1}{2}\log\left(\frac{1-\varepsilon_t}{\varepsilon_t}\right)$	Evaluate the normalization factor exactly.
14	$F_{T(x)} = \sum_{t=1}^{T} \alpha_t h_{t(x)}$	Connect the unrolled product to the final real score.
15	$\ell_{\exp}(y, F(x)) = e^{-yF(x)}$	Relate signed score to a smooth empirical upper bound.

IN-CLASS CHECK

Prompt: Reconstruct the fifteen items in order. Which three roles must remain separate before any algebra begins?

Hint: Look for a population guarantee, an objective-shape fact, and an empirical optimization rule.

Answer:

The order is realizable PAC learning; convex-function background; ERM; weak learner; edge; strong learner; example distribution; exponential loss; weak-hypothesis coefficient; AdaBoost; $D_{t(i)}$; ε_t; α_t; $F_{T(x)}$; and exponential-loss notation. PAC learning supplies a population guarantee, convexity describes objective geometry, and ERM is an empirical rule. None can be silently substituted for either of the others.

For binary labels and binary base hypotheses, Tutorial 6A established

$$D_{t+1}(i) = \frac{D_{t(i)} e^{-\alpha_t y_i h_{t(x_i)}}}{Z_t}, \quad Z_t = \sum_{j=1}^{m} D_{t(j)} e^{-\alpha_t y_j h_{t(x_j)}}, \tag{19}$$

with $D_1(i) = \frac{1}{m}$ and every $Z_t > 0$. The symbol Z_t is the **normalization factor**: the total unnormalized mass created at round t. It is a local proof quantity, not a new course-wide notation export.

Before the theorem is stated, complete the four-rung ladder. In the middle column put either $=$ or $\leq$; in the right column name the fact that licenses the relation. The final line must contain only the known round quantities ε_t or $\gamma_t = \frac{1}{2} - \varepsilon_t$.

Left-hand quantity	**Relation**	**Right-hand quantity / reason**
$\hat{R}_{S(\text{sign}(F_T))}$		$\frac{1}{m}\sum_i e^{-y_i F_{T(x_i)}}$ /
$\frac{1}{m}\sum_i e^{-y_i F_{T(x_i)}}$		$\prod_{t=1}^{T} Z_t$ /
Z_t		$2\sqrt{\varepsilon_{t(1-\varepsilon_t)}}$ /
$\prod_t Z_t$		$\exp\left(-2\sum_t \gamma_t^2\right)$ /

t	ε_t	γ_t	Z_t	cumulative product
1	0.30			
2	0.25			
3	0.40			

IN-CLASS CHECK

Prompt: Commit all four relation symbols before reading the proof. Why are the first and final relations inequalities while the middle two are exact identities?

Hint: Start from the pointwise indicator bound; then ask what normalization preserves exactly.

Answer:

The first relation is $\leq$ by $1_{[y_i F_T(x_i) \leq 0]} \leq e^{-y_i F_{T(x_i)}}$. The second and third relations are equalities: unrolling the normalized update makes the exponential average equal to $\prod_t Z_t$, and the correct/error split gives the exact two-square-root factor. The last relation is $\leq$ and uses $1 - x \leq e^{-x}$ after writing $Z_t = \sqrt{1 - 4\gamma_t^2}$.

The second rung is where the normalization does the real work. Repeatedly substitute the update for one fixed index i:

$$D_{T+1}(i) = \frac{D_1(i)\exp\left(-\sum_{t=1}^{T}\alpha_t y_i h_{t(x_i)}\right)}{\prod_{t=1}^{T} Z_t} = \frac{\exp\left(-y_i F_{T(x_i)}\right)}{m\prod_{t=1}^{T} Z_t}. \tag{20}$$

Rearrange, sum over i, and use $\sum_i D_{T+1}(i) = 1$:

$$\frac{1}{m}\sum_{i=1}^{m}\exp\left(-y_i F_{T(x_i)}\right) = \prod_{t=1}^{T} Z_t. \tag{21}$$

This is an exact identity. It depends on the uniform initial distribution and the normalized AdaBoost update; it is not an approximation and not a population expectation.

IN-CLASS CHECK

Prompt: Suppose the initial weights are a normalized distribution D_1 but are not uniform. What exact quantity replaces the unweighted empirical exponential average?

Hint: Repeat the unrolling without replacing $D_1(i)$ by $\frac{1}{m}$.

Answer:

The identity becomes $\sum_i D_1(i)e^{-y_i F_T(x_i)} = \prod_t Z_t$. The ordinary unweighted empirical average appears specifically because AdaBoost initializes $D_1(i) = \frac{1}{m}$. The round normalization still telescopes exactly, but the starting measure determines the average.

Now split Z_t into the mass classified correctly and incorrectly by h_t. Their D_t-masses are $1 - \varepsilon_t$ and ε_t, so

$$Z_t = (1-\varepsilon_t)e^{-\alpha_t} + \varepsilon_t e^{\alpha_t}. \tag{22}$$

Substituting the registered log-odds coefficient gives

$$Z_t = (1-\varepsilon_t)\sqrt{\frac{\varepsilon_t}{1-\varepsilon_t}} + \varepsilon_t\sqrt{\frac{1-\varepsilon_t}{\varepsilon_t}} = 2\sqrt{\varepsilon_{t(1-\varepsilon_t)}}. \tag{23}$$

This expression is exact when the binary round has $0 < \varepsilon_t < \frac{1}{2}$. With the round edge $\gamma_t = \frac{1}{2} - \varepsilon_t$,

$$Z_t = 2\sqrt{\left(\frac{1}{2}-\gamma_t\right)\left(\frac{1}{2}+\gamma_t\right)} = \sqrt{1-4\gamma_t^2}. \tag{24}$$

Since $0 < 4\gamma_t^2 < 1$, the general inequality $1 - x \le e^{-x}$ implies

$$\sqrt{1-4\gamma_t^2} \le e^{-2\gamma_t^2}. \tag{25}$$

No small-edge approximation has been used.

Result: AdaBoost empirical training-error bound

For binary AdaBoost initialized by $D_1(i) = \frac{1}{m}$, using the registered coefficient and normalized update for T finite rounds with $0 < \varepsilon_t < \frac{1}{2}$,

$$\hat{R}_{S(\mathrm{sign}(F_T))} \le \prod_{t=1}^{T} Z_t = \prod_{t=1}^{T} 2\sqrt{\varepsilon_{t(1-\varepsilon_t)}} \le \exp\left(-2\sum_{t=1}^{T}\gamma_t^2\right), \quad \gamma_t = \frac{1}{2} - \varepsilon_t. \tag{26}$$

If one fixed $\gamma > 0$ satisfies $\gamma_t \ge \gamma$ for every round, then

$$\hat{R}_{S(\mathrm{sign}(F_T))} \le \exp(-2\gamma^2 T). \tag{27}$$

Every right-hand quantity is an upper bound on observed empirical error; it is not the observed error itself and it is not a test-error guarantee.

IN-CLASS CHECK

Prompt: Fill the three-round ledger for $\varepsilon_1 = 0.30$, $\varepsilon_2 = 0.25$, and $\varepsilon_3 = 0.40$. Compute the exact-factor product, the varying-edge bound, and the fixed-edge corollary using the minimum edge.

Hint: The edges are 0.20, 0.25, and 0.10. Specialize to one common edge only after using all three values.

Answer:

The exact factors are $\sqrt{0.84} \approx 0.9165$, $\sqrt{0.75} \approx 0.8660$, and $\sqrt{0.96} \approx 0.9799$. Their product is approximately 0.7778. The varying-edge certificate is $\exp(-2(0.20^2 + 0.25^2 + 0.10^2)) = \exp(-0.225) \approx 0.7985$. The fixed-edge corollary is $\exp(-2(0.10)^2(3)) = \exp(-0.06) \approx 0.9418$. Thus 0.7778 is tightest here, 0.7985 is next, and 0.9418 is loosest. All three are certificates, not observations.

Convexity explains why the one-dimensional coefficient calculation has a well-behaved objective. It does not create the exponential rate by itself. The rate uses a pointwise loss upper bound, the normalized-weight telescoping identity, the exact value of Z_t, an edge condition, and the exact inequality $1 - x \le e^{-x}$.

IN-CLASS CHECK

Prompt: Audit the claim: "AdaBoost's training error decays exponentially because exponential loss is convex." What is missing?

Hint: Name the chain of facts that connects a smooth objective to a zero-one error certificate.

Answer:

Convexity alone is insufficient. The proof also needs the indicator upper bound, the exact telescoping identity produced by normalized updates, the exact formula $Z_t = 2\sqrt{\varepsilon_{t(1-\varepsilon_t)}}$, a positive edge at the relevant rounds, and $1 - x \le e^{-x}$. Even the completed empirical theorem says nothing by itself about population or test error.

The theorem can drive its upper bound toward zero. Once every training sign is correct, however, the zero-one count stops recording how far each real score lies from zero. That lost information is the next object to inspect.

Scheduled 15-minute break

Return able to reconstruct the chain $\hat{R}_S \le \prod_t Z_t \le \exp\left(-2\sum_t \gamma_t^2\right)$ without looking back. The break introduces no new required knowledge.

Read what zero empirical error leaves invisible

The two panels below are intentionally unnamed. Each lists the signed training scores $y_i F_{T(x_i)}$ for eight examples at a round when every sign is correct. Mark the decision threshold, compute the training error in each panel, and then circle the scores that would change sign after a downward perturbation of 0.15.

Panel A signed scores	Panel B signed scores
$0.04, 0.08, 0.12, 0.24$	$0.38, 0.51, 0.63, 0.78$
$0.35, 0.61, 0.90, 1.20$	$0.92, 1.05, 1.31, 1.55$
training error:	training error:
affected by -0.15:	affected by -0.15:

Later validation trace	can decrease?	can flatten?	can increase?
Allowed by zero training error alone			

Table 3: Two positive signed-score distributions and three candidate validation behaviors. Zero training error fixes the signs, not these magnitudes or the validation trace.

IN-CLASS CHECK

Prompt: What does the common zero-error count hide, and which of the three validation behaviors can zero training error alone rule out?

Hint: Separate information measured on the training sample from behavior on untouched data.

Answer:

> Both panels have empirical error zero because all eight signed scores are positive. Panel A has three scores below 0.15 and Panel B has none, so the same count hides substantial differences in score magnitude and distribution. Zero training error alone rules out none of the three validation traces: validation error may decrease, flatten, or increase.

Additional valid AdaBoost rounds can change F_T, so they can move the signed training scores even after their signs are all positive. On some benchmark problems, test error has continued to improve long after training error reached zero. That is a **possible empirical pattern**, not a universal theorem. It does not imply monotone improvement, a fixed optimal round count, or a rule that one should always continue boosting.

Result: A qualified post-zero diagnosis

Zero empirical classification error establishes only $y_i F_{T(x_i)} > 0$ for every training example under the declared sign convention. It does not determine the magnitudes or distribution of these signed scores. Later rounds may change those quantities, and test performance may improve, but any statement about test behavior requires separate evidence. Noise, persistent hard examples, an insufficient base-hypothesis class, and repeated use of validation data can all change the outcome.

IN-CLASS CHECK

Prompt: A run reaches zero training error at round 12, and validation error falls from 9% to 8% by round 30. Which conclusion is supported: "continue forever," "stop at 12," or a narrower statement?

Hint: Report only the observed interval and keep future rounds and new data outside the claim.

Answer:

> The supported statement is narrower: on this run and this untouched validation evaluation, continuing from round 12 to round 30 accompanied a one-point validation-error decrease. The observation proves neither monotone future improvement nor that round 30 is optimal. A stopping or selection rule must declare how validation data are used and must preserve an untouched final evaluation layer.

IN-CLASS CHECK

Prompt: Why can persistent noisy or mislabeled examples make "later rounds improve confidence" an unsafe universal explanation?

Hint: Ask which examples exponential weighting keeps emphasizing and whether their labels are learnable.

Answer:

> AdaBoost increases relative attention on examples that remain hard. If a label is wrong or the base class cannot represent the needed distinction, later rounds may concentrate effort on a persistent conflict rather than improving useful prediction confidence. The signed-score distribution can still change, but its direction and its relationship to test performance are data- and model-dependent.

Chapter 7 will formally define a normalized confidence quantity and derive a generalization theorem that uses its empirical distribution together with the base-class complexity. Those objects are not available in this core Tutorial. Here we use only the already defined signed score $y_i F_{T(x_i)}$ and ordinary comparisons of its values.

The two panels have exposed a second boundary: an algorithmic update, a loss decrease, and a change in signed scores can occur together, yet each observation answers a different question. We now align them without merging their claims.

Match the view to the question it can answer

For one AdaBoost round, fill the shared-state spine first: D_t, h_t, ε_t, α_t, D_{t+1}, and the updated score $F_t = F_{t-1} + \alpha_t h_t$. Then assign each diagnosis card to one column and cross out the inference that the column does not support.

Prompt	Column A	Column B	Column C
quantity to inspect			
supported question	Which examples will matter more next round?	Why is this coefficient chosen?	What changed after the signs became correct?
candidate overclaim	Therefore test error falls.	Therefore convexity proves generalization.	Therefore every later round improves test error.
cross out and repair			

IN-CLASS CHECK

Prompt: Assign the three columns to algorithmic mechanics, roundwise optimization, and signed-score evolution. Name the shared quantity that connects each adjacent pair.

Hint: The update links weights to the selected hypothesis; the coefficient links the loss step to the new ensemble score.

Answer:

Column A is the algorithmic view and inspects D_t and D_{t+1}. Column B is the roundwise optimization view and inspects the exponential objective, ε_t, and its exact minimizing coefficient α_t. Column C is the confidence-evolution view and inspects signed scores $y_i F_{t(x_i)}$. The selected h_t and weighted error connect A to B; $\alpha_t h_t$ connects B to the update of F_t in C.

The three readings are complementary:

- The **algorithmic view** tracks the normalized training-index distribution. It explains which examples influence the next base-hypothesis call and verifies that the implementation remains a distribution.
- The **roundwise optimization view** tracks the exact one-dimensional decrease of empirical exponential loss. It explains the log-odds coefficient and, together with normalization and edge conditions, contributes to the empirical training bound.
- The **confidence-evolution view** tracks how the signed scores $y_i F_{t(x_i)}$ change across examples. It explains what the zero-one training count discards and motivates a later generalization analysis.

Result: Three complementary ways to read an AdaBoost round

The views share AdaBoost quantities, but they license different claims. Reweighting supports an implementation statement; exact line minimization supports an empirical objective statement; confidence evolution supports a diagnosis of information hidden by zero-one error. None of these statements, alone or by renaming, is a population generalization theorem.

IN-CLASS CHECK

Prompt: Choose the useful view and reject one unsupported inference for each task: (a) debug weights that do not sum to one; (b) derive α_t; (c) compare two zero-error ensembles; (d) claim low future test error.

Hint: The fourth task cannot be completed from any single Chapter 6 view.

Answer:

> Use the algorithmic view for (a), the roundwise optimization view for (b), and confidence evolution for (c). Task (d) is unsupported by all three in Chapter 6. It needs a separate generalization result with declared complexity and sampling conditions, plus honest evaluation for an empirical test claim.

There is a formal theorem, under a finite base-hypothesis representation and a specified greedy direction rule, that relates AdaBoost to coordinate descent on empirical exponential loss. We have not stated its definition, assumptions, or equivalence proof here. Chapter 7 reconstructs them before use. Likewise, Chapter 7 defines the normalized confidence quantity before stating any distributional or generalization bound. Neither future theorem is a hidden premise of the three-view interface.

IN-CLASS CHECK

Prompt: Explain why "the three views are equivalent" is too strong even though they share D_t, α_t, and F_t.

Hint: Distinguish a shared state description from an interchangeable guarantee.

Answer:

> Shared quantities allow translations between descriptions of one AdaBoost round, but the conclusions have different logical types. A normalized update is an algorithmic invariant; a line minimum is an empirical optimization fact; a signed-score shift is a descriptive observation. None automatically supplies the assumptions or conclusion of a population theorem, so their guarantees are not interchangeable.

Standard language to carry forward

Term or notation	Meaning here	Recall cue
normalization factor	The total unnormalized example mass produced at round t.	$Z_t = \sum_i D_{t(i)} e^{-\alpha_t y_i h_{t(x_i)}}$
training-error product bound	The empirical zero-one error is no larger than the product of exact round factors.	$\hat{R}_S \le \prod_t Z_t$
varying-edge bound	Each round contributes its own squared edge to an exponential certificate.	$\exp\left(-2\sum_t \gamma_t^2\right)$
fixed-edge corollary	One common positive lower bound on every round edge yields exponential decay in T.	$\exp(-2\gamma^2 T)$
algorithmic view	Tracks normalized example weights and the next learner call.	$D_t \to D_{t+1}$
roundwise optimization view	Tracks exact one-dimensional empirical exponential-loss reduction.	$\varepsilon_t \to \alpha_t$
confidence-evolution view	Tracks signed-score information discarded by the sign count.	$y_i F_{t(x_i)}$

Reconstruct the exact Chapter 7 entry

Chapter 7 may use exactly four items from this boundary, in order. VC dimension is a verified pass-through item from Chapter 2, not a Chapter 6 introduction.

No.	Available item	What Chapter 7 may ask you to do
1	AdaBoost	Execute and interpret the normalized update and real ensemble score.
2	AdaBoost training-error product bound	Reconstruct the exact product, varying-edge, and fixed-edge chain.
3	VC dimension	Supply the established base-class capacity notion for a later generalization term.
4	three complementary AdaBoost views	Select the view that matches a claim and state what it does not prove.

IN-CLASS CHECK

Prompt: Without looking back, reconstruct the four Chapter 7 entry items in order. Why are the optional score normalization below and the coordinate-descent equivalence absent?

Hint: Separate established Chapter 6 exports from future definitions and theorems.

Answer:

> The order is AdaBoost; the AdaBoost training-error product bound; VC dimension; and the three complementary AdaBoost views. The optional comparison below is zero-minute, unassessed, and non-exported. Formal normalized confidence and the coordinate-descent equivalence require new Chapter 7 definitions, assumptions, and proofs, so they cannot be boundary inputs.

Source note

The historical Week 6 Learning Sheet supplies the official Tutorial 6B scope on printed/PDF pp. 21-51. Its bound-like numbers are not treated as observed errors, its approximation narrative is replaced by the exact inequality $1 - x \leq e^{-x}$, and its universal post-zero and view-equivalence suggestions are not retained.

The indicator-to-product proof, exact normalization factor, varying-edge bound, and fixed-edge corollary follow Mohri, Rostamizadeh, and Talwalkar, Theorem 7.2 and its proof, printed pp. 149-150 / PDF pp. 166-167, checked also against Shalev-Shwartz and Ben-David, Section 10.2, printed/PDF pp. 135-137. Schapire, **Explaining AdaBoost**, Sections 3-5, printed/PDF pp. 4-11, supplies the post-zero empirical observation and the complementary explanations, together with the warning that exponential-loss minimization alone does not guarantee good generalization. Its noise and insufficient-expressiveness discussion on pp. 12-14 supplies the qualification boundary.

Formal normalized confidence, its empirical distribution and generalization theorem, and the coordinate-descent equivalence are reserved for Chapter 7.

Optional self-study appendix

OPTIONAL DEEP DIVE - 0 CLASS MINUTES

Question: Can two zero-error ensembles be compared after rescaling each one's signed scores to a common coefficient scale, without importing Chapter 7's formal definition?

Answer.

Yes, as a descriptive comparison only. For each ensemble, divide all of its signed training scores by that ensemble's same positive total coefficient weight. Then compare how much empirical mass lies near zero and how the rescaled values shift between rounds. A rightward shift can reveal changes hidden by the zero-one count.

This calculation does not establish the registered Chapter 7 normalized confidence definition, a reusable distribution object, or a generalization theorem. It gives no monotone test-error guarantee and exports no knowledge. Chapter 7 rebuilds the formal object, states its scale invariance and range, and adds the required complexity and sampling conditions before any bound is used.

Post-class or self-study. Not a prerequisite for the core path.